

JOPLIN 24 N, MO, USA

WMO: 725590

Lat: 37.4277N

Lon: 94.5829W

Elev: 952

StdP: 14.20

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 23908

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.4	12.2	-1.4	5.3	8.6	3.2	6.7	13.5	18.0	51.8	16.2	53.9	5.0	N/A	0.406

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.4	95.9	74.5	92.5	75.2	89.9	75.3	78.7	89.4	77.7	88.3	76.6	87.0	6.2	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.8	139.8	85.3	74.6	134.0	84.2	73.4	128.5	82.9	43.0	89.4	41.7	88.4	40.6	87.0	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.1	13.9	12.1	-0.4	98.4	7.2	5.1	-5.6	102.0	-9.8	105.0	-13.8	107.8	-19.0	111.5		
			-1.8	80.7	7.4	1.2	-7.1	81.5	-11.5	82.2	-15.6	82.9	-21.0	83.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.9	33.2	36.8	47.9	57.1	65.7	75.1	78.1	76.5	69.6	57.7	46.5	36.8
	DBStd	18.04	11.62	11.90	11.05	8.92	7.86	5.05	5.10	5.68	6.82	9.22	9.88	9.88
	HDD50	1763	531	395	171	34	3	0	0	0	0	32	175	422
	HDD65	4398	986	791	537	262	92	3	1	1	31	262	558	874
	CDD50	4265	10	24	106	249	489	753	870	821	589	272	69	13
	CDD65	1424	0	0	6	26	113	306	406	357	170	37	3	0
	CDH74	13884	0	7	53	207	921	2912	4225	3641	1570	333	15	0
	CDH80	5702	0	0	5	22	213	1102	1976	1728	588	67	1	0
Wind	WSAvg	4.7	5.5	6.0	6.6	6.7	5.0	4.7	3.8	3.0	2.9	3.5	4.5	4.7
Precipitation	PrecAvg	47.7	1.5	1.9	3.4	5.0	6.8	6.4	4.4	3.1	5.0	3.5	3.3	2.5
	PrecMax	63.3	3.2	5.5	8.5	12.9	13.9	16.5	15.9	7.1	17.9	9.6	9.7	8.1
	PrecMin	30.4	0.3	0.0	1.4	1.1	1.6	1.5	0.4	0.0	1.3	0.7	0.0	0.4
	PrecStd	8.9	1.0	1.4	1.9	2.6	3.1	3.1	2.9	1.7	3.5	2.2	2.5	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.3	74.2	79.8	82.7	87.4	94.3	100.4	101.5	93.5	85.8	76.1	68.5
		MCWB	57.3	59.6	63.9	64.9	71.4	76.0	73.5	75.5	73.2	71.2	61.1	60.2
	2%	DB	62.1	66.9	74.3	79.6	85.0	90.7	96.1	96.2	89.6	81.8	70.9	62.9
		MCWB	51.0	56.5	61.0	63.6	70.6	76.0	74.2	75.1	74.7	66.1	59.0	55.3
	5%	DB	57.3	62.0	69.6	76.3	82.4	88.6	92.7	92.7	86.8	77.8	67.1	57.6
		MCWB	48.5	54.1	57.0	62.6	69.6	75.1	75.9	73.9	73.2	64.2	57.7	49.7
	10%	DB	51.2	56.6	65.7	72.6	79.7	86.6	89.8	89.4	83.3	73.7	62.9	53.2
		MCWB	44.5	48.2	54.8	61.1	68.3	74.4	76.1	74.5	71.1	62.7	54.1	47.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.4	62.4	66.1	69.4	75.0	79.3	80.5	80.1	77.4	72.9	65.4	61.6
		MCDB	67.0	70.5	77.5	79.2	84.2	90.3	90.6	90.3	89.0	83.1	70.0	67.3
	2%	WB	52.8	59.2	63.0	66.8	73.2	77.9	79.0	78.4	75.7	70.5	61.5	56.4
		MCDB	59.0	65.2	72.6	75.4	82.3	88.4	89.3	89.7	87.4	77.7	67.3	60.6
	5%	WB	48.9	55.0	60.0	64.8	71.4	76.6	77.9	77.1	74.3	67.9	58.7	51.7
		MCDB	55.6	60.9	67.5	73.2	79.9	86.8	88.2	88.2	85.0	73.6	65.1	55.7
	10%	WB	44.2	48.0	56.7	62.7	69.7	75.3	76.8	75.5	72.8	64.8	55.4	47.6
		MCDB	51.0	56.6	63.1	70.5	77.7	84.9	86.9	85.8	81.9	71.7	61.7	52.5
Mean Daily Temperature Range	5% DB	MDBR	20.5	20.4	21.5	21.3	19.5	19.2	20.4	21.6	22.1	23.4	22.5	19.1
		MCDBR	28.2	28.5	26.4	24.7	22.6	21.1	24.5	26.5	24.3	26.4	26.6	26.1
		MCWBR	18.9	19.3	14.9	13.6	11.2	9.7	8.5	9.4	10.8	13.4	17.2	17.2
	5% WB	MCDBR	25.1	26.7	21.8	20.4	19.8	20.0	19.8	21.8	22.0	20.0	23.3	21.1
		MCWBR	18.7	20.2	13.8	12.9	10.8	10.0	9.1	9.7	10.3	12.0	17.3	16.7
Clear-Sky Solar Irradiance	taub		0.303	0.313	0.345	0.380	0.422	0.424	0.439	0.441	0.387	0.345	0.310	0.290
	taud		2.449	2.432	2.389	2.313	2.253	2.281	2.279	2.251	2.370	2.464	2.504	2.507
	Ebn at Noon		281	292	291	285	274	272	267	264	274	277	277	279
	Edh at Noon		26	30	35	40	43	42	42	42	36	29	25	23
All-Sky Solar Radiation	RadAvg		758	949	1258	1600	1797	2066	2054	1843	1556	1133	807	645
	RadStd		68	145	82	110	148	166	128	118	115	133	89	73

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page